Predicting County-Level Diabetes Risk: A Health Data Science Approach Using Social Determinants of Health

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Introduction:

Type 2 diabetes is undoubtedly a health concern of great public health significance in America, with more than thirty-eight million adults afflicted by this disease, leading to considerable morbidity and mortality rates from heart disease, kidney failure, neuropathy, and premature death (1). The incidence of type 2 diabetes has also seen sharp increase in recent decades, with an estimate from the CDC suggesting that 12 to 14 percent of adults in America are afflicted by this disease (2). While there is no doubt that genetic factors and personal lifestyle factors are important in this regard, recent studies emphasize the role of SDOH factors such as income inequality, education levels, access to healthcare facilities, housing conditions, food security, and community environments in type 2 diabetes (3, 4, 5). Such factors are said to have a greater influence on type 2 diabetes than other factors (5).

Type 2 diabetes has significant financial cost. Over $300 billion spent on medical care each year, and an additional $90 billion spent on disability and lost productivity (5). The significance of switching from reactive clinical care to proactive, population level initiatives that address structural determinants of health prior to illness development is shown by these startling statistics.

For precisely this type of initiative, the CDC's Health Impact in 5 Years (HI-5) framework offers a strategic organizational framework. In three preventive buckets, traditional clinical prevention, novel clinical treatments, and community-wide initiatives that address socioeconomic determinants, it finds evidence-based policies and programs that produce quantifiable benefits in population health within five years (6). This project fits neatly inside this structure.

Research has repeatedly shown that socioeconomic factors at the county level, such as rurality, insurance coverage, educational attainment, and child poverty, are among the best indicators of diabetes prevalence at the population level (3, 7). However, rather than employing precise county-level targeting, the majority of public health initiatives distribute resources using broad, state-level estimations. Due to this disparity, many of the most burdened areas do not receive adequate assistance.

This project's goals are to identify high-risk counties for type 2 diabetes nationwide using county-level data and machine learning techniques, investigate the underlying structural factors influencing risk, and suggest a data-driven, equity-focused public health intervention that is in line with the CDC's HI-5 framework. Public health professionals can improve health outcomes and significantly reduce regional and socioeconomic health inequalities by focusing on high-risk groups and their systemic causes (6).

Literature Review:

Over the past 20 years, there has been a growing recognition of the importance that socioeconomic determinants of health play in managing chronic diseases, including type 2 diabetes. Eight peer reviewed sources that influenced our choice of variables, modeling strategy, and suggested actions are summarized in the review that follows.

According to Hill et al. (3), diabetes prevalence is predicted by neighborhood-level socioeconomic variables, such as income, educational attainment, housing stability, and work prospects, regardless of individual lifestyle or genetic factors. Their socioecological paradigm emphasizes the necessity of taking into account the larger environmental and social settings that influence health outcomes in addition to individual-focused therapeutic interventions. For instance, despite individual lifestyle choices, those who live in low-income communities with restricted access to wholesome food or secure recreational areas are significantly more likely to acquire diabetes. Our choice of SDOH variables as the only feature set for modeling diabetes risk at county level is explicitly supported by this approach.

This strategy was verified at individual level by Walker, Smalls, and Egede (11), who showed that social determinants are independently linked to glycemic control even after adjusting for clinical factors like medication adherence and BMI. Using SDOH-only features our machine learning models was directly supported by their discovery that SDOH variables alone had substantial predictive value for diabetes outcomes.

The epidemiological background for this investigation was established by Menke et al. (2), who used nationally representative data from 1988 to 2012 to illustrate the long-term increase trend in U.S. diabetes prevalence. Their discovery that between 12 and 14% of American people satisfied diagnostic criteria validates the scope of the issue and puts our 10.4% prevalence cutoff point for high-risk regions context.

In a longitudinal research, Berkowitz et al. (4) looked at how food insecurity affected glycemic control. Their research showed that patients had worse diabetes management results and higher HbA1c levels when they lived in places with restricted access to reasonably priced, nutrient-dense food. The results highlight the importance of environmental and socioeconomic factors in diabetes risk and the need for community-level interventions such as mobile markets, subsidized groceries, and nutrition education initiatives. Our HI-5 Bucket 1 recommendations, which focus on food access high-risk countries, directly supported by this source.

Kolak et al. (7) mapped neighborhood-level variables linked to diabetes prevalence throughout the continental United States using high-resolution spatial data. Clusters of high-risk counties with socioeconomic disadvantages, like poverty, poor educational attainment, and inadequate healthcare infrastructure, were found by their investigation. The study supports creation of our interactive surveillance dashboard and validates our FIPS-based county-level methodology by highlighting significance of geospatial analysis in identifying priority zones for intervention and resource allocation.

According to Meit et al. (8), compared to their urban counterparts, rural Americans have much higher rates of chronic illness, less insurance coverage, and significantly less access to preventative treatment. This source strongly supports our advice to implement telehealth infrastructure and mobile health clinics

in high-rurality counties, and it is especially pertinent to our discovery that the percentage of population living in rural areas was third highest predictor of high diabetes risk in our Random Forest model.

The connection between psychosocial stresses and metabolic dysregulation is covered by Egede and Walker (10). Chronic stress brought on by prejudice, social isolation, or unstable finances can raise cortisol levels, which can lead to insulin resistance and poor glucose metabolism. The significance of economic stability initiatives (SNAP outreach, EITC programs) as indirect diabetes preventive methods reinforced by this biological mechanism by which socioeconomic factors transfer into clinical illness.

Walker et al. (11) discovered that diabetes prevalence and complication rates were higher in regions with greater uninsured rates and fewer primary care physicians. Screening, early detection, and illness treatment are routinely improved by increasing access through Medicaid outreach and federally qualified health centers (FQHCs), especially in underprivileged groups. Our Bucket 3 clinical and social services initiatives are directly informed by this source.

The hypothesis that SDOH variables are both necessary and sufficient to predict diabetes risk at the county level and that structural interventions addressing these determinants can yield quantifiable improvements in population health outcomes within the five year window specified by HI-5 framework is supported by strong body of evidence.

Dataset Description:

In order to assure accurate geographic linkage between datasets, this research combined county-level data from three publicly accessible government sources using standardized five-digit government Information Processing Standards (FIPS) codes.

Primary Data Sources:

The main outcome variable, county-level diagnosed diabetes prevalence reported as the proportion of individuals with diabetes diagnosis, was obtained from the CDC Diabetes Surveillance System (2023). The 2023 estimates in this dataset, which includes 3,143 U.S. counties and county equivalents, were obtained using small-area estimating techniques and the Behavioral Risk Factor Surveillance System.

Nine SDOH predictor variables were available at county level in County Health Rankings and Roadmaps dataset (2023), which was released by the University of Wisconsin Population Health Institute in collaboration with the Robert Wood Johnson Foundation. These variables were chosen because they have theoretical foundation in literature and are available in large enough number of counties to enable reliable modeling.

A third source of data was the five-year estimates from the American Community Survey (ACS). But when CDC and County Health Rankings datasets were combined, it was clear that County Health Rankings already included the major ACS-derived factors associated with diabetes risk, such as median family income, unemployment rates, and housing cost burden. Without providing distinctive predictive

information, integrating raw ACS variables individually would have resulted in significant multicollinearity (variance inflation factors greater than 30). As a result, ACS variables were not included as stand-alone source; instead, the CHR dataset captures their impact. Future research should examine ACS-specific characteristics including housing tenure and comprehensive food insecurity indexes that are not included in CHR.

Variable:

Counties in the top 25th percentile of adult diabetes prevalence (at or above 10.4 percent) were categorized as high-risk (1), while all other counties were classed as low-risk (0). This criterion was chosen because it maintains a manageable class imbalance for modeling while representing a level of load that is clinically and epidemiologically significant.

Nine characteristics of SDOH were included as independent variables:

Variable	Source	Direction of Risk
Premature Death Rate (per 100,000)	CHR 2023	Higher → Greater Risk
% Rural Population	CHR 2023	Higher → Greater Risk
% Female	CHR 2023	Higher → Mixed
Uninsured Rate (%)	CHR 2023	Higher → Greater Risk
% Some College	CHR 2023	Higher → Protective
Unemployment Rate (%)	CHR 2023	Higher → Greater Risk
Median Household Income ($)	CHR 2023	Lower → Greater Risk
Children in Poverty (%)	CHR 2023	Higher → Greater Risk
Severe Housing Cost Burden (%)	CHR 2023	Higher → Greater Risk

Preprocessing and Merging:

A final analytic dataset of 2,879 ranked counties was created by combining the CDC and CHR information on FIPS codes. Since CHR cannot generate accurate SDOH estimates for unranked counties, which are primarily independent cities in Virginia and their counterparts in other states, they were not included. There was little data loss since rows with missing values in any feature were eliminated. Before modeling, all continuous characteristics were normalized using z-score normalization to guarantee comparability across various scales and units.

Descriptive statistics, Pearson correlation matrices, and comparison box plots stratified by diabetes risk class were all used in the exploratory data analysis. Across counties, the median household income varied from around $28,000 to $125,000. Rurality ranged from 0 to 100 percent, while childhood poverty rates

ranged from 5 to 38%. The strong structural patterning of diabetes risk was confirmed by fact that counties in the highest-risk quartile had much greater rates of poverty and early death, worse educational attainment, and significantly less access to healthcare than low-risk counties.

Data Analysis and Results:

Features Selection using LASSO Regularization

All nine candidate characteristics underwent a variance inflation factor study before the model was trained. The VIF scores, which varied from 1.8 to 4.9, showed controllable multicollinearity that was much below the traditional cutoff point of ten. All nine traits were kept for further examination, and none were removed based only on VIF.

Data-driven feature selection was performed using LASSO regularization (LassoCV with five-fold cross-validation) on continuous diabetes prevalence outcome. All nine SDOH characteristics had non-zero coefficients assigned by LASSO, indicating that each predictor provided significant independent information. Cross-validation was used to determine ideal regularization strength (alpha), and obtained coefficient magnitudes informed the interpretation of relative feature significance.

Logistic Regression Baseline

An 80/20 stratified train-test split was used to train logistic regression classifier with balanced class weights. Test set accuracy of 73.8%, AUC-ROC of 0.791, precision of 50.0%, recall of 72.0%, and cross-validated AUC of 0.806 were all attained by model. Higher educational attainment (% some college) and median family income were greatest protective factors, while child poverty and early mortality rates were strongest positive predictors of high-risk categorization, according to the interpretable odds ratios. The theoretical expectations from the literature study are in close agreement with these findings.

Metric	Logistic Regression	Random Forest
Accuracy	73.8%	80.7%
AUC-ROC	0.791	0.829
Precision	50.0%	66.0%
Recall	72.0%	55.0%
CV AUC (5-fold)	0.806	0.827

Model performance is compared in above table. With the exception of recall, where Logistic Regression found more actual high-risk counties at the expense of more false positives, Random Forest beat Logistic Regression in every parameter. The increased recall of logistic regression may be useful for public health screening applications where false negatives are more expensive than false positives. However, Random Forest is best model for resource prioritization and risk assessment because of its better AUC and accuracy.

Random Forest Modeling

Non-linear interactions between SDOH characteristics that the logistic regression model was unable to capture were caught using Random Forest classifier (200 estimators, maximum depth of 10, balanced class weights). With accuracy of 80.7% and AUC of 0.829, the model demonstrated no overfitting, as confirmed by cross-validated AUC of 0.827.

The following ranking of SDOH predictors was found using feature significance analysis utilizing mean decrease in Gini impurity:

Rank	SDOH Predictor	Importance (%)
1	Premature Death Rate	31%
2	Children in Poverty	22%
3	% Rural Population	16%
4	Uninsured Rate	13%
5	Median Household Income	10%
6	% Female	3%
7	Unemployment Rate	2%
8	Some College (%)	2%
9	Severe Housing Cost Burden	1%

With 31% of overall feature significance, the premature death rate was by far the most significant predictor. This variable is potent proxy for the combined effect of several SDOH determinants on population health as it represents the accumulated chronic illness burden in community. The second and third most significant characteristics were the percentage of children living in poverty (22%) and the percentage of children living in rural areas (16%), indicating the importance of economic hardship and geographic isolation in determining the risk of diabetes.

The top three predictors were agreed upon by both models. In line with the socioecological concept of Hill et al. (3), interaction effects found by Random Forest showed compounding risk when low educational attainment combined with food insecurity or high poverty. 753 U.S. counties, mostly in deep South, Appalachia, the Mississippi Delta, and portions of the rural West, were classified as high-risk by the model.

Geographic Distribution

Strong regional clustering was seen when projected risk scores were geographically analyzed. Mississippi, West Virginia, Alabama, Kentucky, and Louisiana had the largest percentages of high-risk counties. These states are known for their high rates of poverty, rural isolation, and inadequate healthcare infrastructure.

This trend highlights the necessity for locally focused rather than uniform national treatments and is consistent with the socioeconomic and geographic patterning of diabetes reported in literature (7).

County-level public health officials and state legislators can use interactive choropleth map created in the analysis dashboard to pinpoint their particular counties, look at the SDOH drivers that are most likely to increase risk, and compare model predictions with local expertise to guide resource allocation decisions.

Proposed Intervention:

The results of machine learning study serve as sole foundation for the suggested action. Each of the suggestions listed below is closely related to characteristic that both models found to be significant predictor, guaranteeing that the intervention is data-driven rather than opinion-based. The concept is based on the CDC HI-5 three-bucket paradigm, which aims to attain quantifiable population health effect in five years (6).

Target audiences include: (1) public health officials at the county level who are in charge of allocating resources and implementing programs; (2) federal and state legislators who have an impact on funding sources, Medicaid policy, and SNAP administration; and (3) FQHC clinicians, community health workers, and social service providers who carry out interventions locally.

Bucket 1: Healthy and Safe Environments

Structural environmental constraints that raise the risk of diabetes are identified by Rural Population predictor (driver #3, 16 percent feature relevance) and Severe Housing Cost Burden (driver #9). Among suggested interventions are:

- In 753 high-risk counties with high rurality ratings, implement mobile health clinics and increase telehealth infrastructure. Mobile clinics have been found to increase diabetes screening and management rates in rural regions and to be successful in reaching populations in places where there is scarcity of providers.
- In counties where food insecurity and housing costs coexist, fund SNAP outreach programs and increase food access efforts like community gardens and mobile markets. Food availability and glycemic outcomes have been shown to be directly correlated by Berkowitz et al. (4), and these treatments target the structural cause of that correlation.
- To lessen the financial strain that causes metabolic dysregulation through cortisol elevation, counties with greatest severe housing cost burden should offer utility assistance and affordable housing vouchers (10).

Bucket 2: Reduce Risky Behaviors

Economic insecurity increases risk of diabetes through behavioral and structural routes identified by Children in Poverty predictor (driver 2, 22 % relevance), the Unemployment Rate (driver 7), and the Some College proportion (driver 8). Among the suggested interventions are:

- In counties with high rates of childhood poverty, implement the CDC National Diabetes Prevention Program (DPP). In high-risk groups, the DPP is systematic lifestyle intervention that has been shown to be successful in lowering the incidence of diabetes by 58% (12). The initiative will reach individuals who are most likely to benefit if high-poverty counties are given priority.
- To lessen the economic instability that contributes to bad health behaviors and restricts access to healthcare, employer-sponsored health insurance should be used to fund workforce health programs and job training initiatives in high-unemployment regions.
- In counties with low college enrollment rates, fund outreach programs for diabetes education and community health literacy. In order to improve health decision-making, community-based health literacy initiatives can replace formal educational attainment. Education is protective factor against diabetes.

Bucket 3: Clinical and Social Services

Gaps in clinical and social safety net coverage are main causes of high-risk categorization, according to Premature Death Rate (driver 1, 31% relevance), Uninsured Rate (driver 4, 13%), and Median Household Income (driver 5, 10%). Among the suggested interventions are:

- In counties with high percentages of uninsured people and early death, increase Medicaid coverage and outreach. According to Walker et al. (11), increasing insurance coverage raises the rates of diabetes management and screening. Medicaid expansion has been linked to significantly lower percentages of uninsured people and better access to preventive care for those who were previously uninsured.
- In counties where primary care capacity is low, increase FQHC funding and staffing. FQHCs are positioned to provide integrated diabetes preventive and management services and act as safety-net providers for populations with inadequate and no insurance.
- In 753 identified high-risk counties, assign community health workers with targeted subsidies to offer risk factor counseling, home-based diabetes education, and social service navigation. When it comes to accessing marginalized communities and lowering obstacles to care, community health professionals have proven to be beneficial.
- In order to improve economic stability in low-income areas, expand EITC outreach and financial literacy initiatives. This would address the income-health route that has been established in this review's many sources.

Data Visualization and Communication

An interactive surveillance dashboard created using Plotly that shows county-level estimated risk ratings on choropleth map of continental United States will be used to disseminate findings. Stakeholders will be able to examine the main contributing characteristics for each chosen county, filter by state, risk tier, or particular SDOH driver, and monitor the implementation of interventions and their results over time. The major target audience for this visual communication approach is state legislators and county public health authorities, who need easily accessible information rather than technical model results.

Data Engineering Plan:

For ongoing public health monitoring, a single cross-sectional analysis is inadequate. The conceptual data pipeline described below allows for yearly updates of county risk model, guaranteeing that intervention targeting stays up to date as social and economic circumstances change.

Pipeline Architecture:

Every April, the CDC and County Health Rankings release their updated datasets, which starts off the pipeline's five consecutive stages:

- Stage 1: Data Ingestion: Updated CSVs are retrieved by automated Python scripts from the County Health Rankings download portal and the CDC Diabetes Surveillance API. To identify newly formed or dissolved counties, FIPS codes are verified against the official Census FIPS master file. Counties that are missing are noted for manual examination.
- Stage 2: Cleaning and FIPS Merge: All datasets are combined based on five-digit FIPS code, raw datasets are cleaned for formatting errors, and missing values are imputed using county-level medians from previous year when feasible. The number of counties with complete vs incomplete data is recorded in completeness report.
- Stage 3: Feature Engineering: Z-score normalization is used to standardize all features. Every year, LassoCV is rerun to either verify that same nine variables are still most informative or to identify any changes in socioeconomic conditions that call for inclusion of additional features.
- Stage 4: Model Re-Scoring: Using data from the previous year, the trained Random Forest model is used to provide updated risk ratings (0–1) for each of the 2,879+ counties. Counties are marked for priority outreach when their risk category shifts from low to high. Every three years, or right away if the cross-validated AUC falls below 0.75, the model is completely retrained.
- Stage 5: Outputs: The real-time surveillance dashboard is updated with risk scores. State health agencies get an annual PDF report that summarizes tier changes and risk profiles at state level. When county's risk tier changes, county health authorities are notified via email notifications. For the aim of allocating federal resources, a ranked intervention priority list of the top 100 highest risk counties is released.

Even when county names vary between dataset vintages, FIPS codes provide constant county-level linking by acting as stable geographic primary key across all data sources and all years. Every run of pipeline is version-controlled, auditable, and reproducible thanks to its automation using GitHub Actions, which are initiated by regular cron job. These characteristics are crucial for public health infrastructure.

CDC Population Health Competencies Addressed

The following CDC Population Health Competencies 1 through 7 are specifically addressed in this project:

Competency	How This Project Addresses It

1. Assess health status using surveillance data	County-level diabetes prevalence assessed using CDC Diabetes Surveillance System and County Health Rankings both federal public health surveillance datasets, across 2,879 counties.
2. Discuss SDOH and environmental determinants	All 9 model features are SDOH variables (poverty, education, housing, income, rurality, insurance) modeled as predictors of diabetes risk at the population level.
3. Integrate biologic/genetic risk with population-level factors	Premature death rate serves as proxy for cumulative chronic disease burden including genetic predisposition, integrated with SDOH predictors in both models.
4. Appraise quality of peer-reviewed evidence	Eight peer-reviewed sources evaluated and synthesized to justify variable selection, modeling approach, and proposed interventions.
5. Apply prevention strategies for populations	All proposed interventions (CDC DPP, SNAP outreach, mobile clinics, Medicaid expansion) are evidence-based primary and secondary prevention strategies.
6. Identify community assets and resources	Intervention matrix identifies FQHCs, SNAP, ACA marketplace navigators, housing authorities, and workforce agencies as community resources for high-risk counties.
7. Community-engagement strategies to reduce disparities	Interventions target underserved populations (uninsured, rural, low-income, high child-poverty counties) using community health workers and participatory outreach to reduce diabetes disparities.

Conclusion:

The distribution of type 2 diabetes among U.S. counties is geographically and socioeconomically structured pandemic that reflects the profound structural injustices ingrained in social, economic, and physical contexts in which Americans live. With our Random Forest model attaining an AUC of 0.829 and our Logistic Regression achieving 0.791 across 2,879 U.S. counties, our experiment showed that Social Determinants of Health alone, without any clinical or genetic variables, can substantially predict county level diabetes risk.

The analysis revealed that there were 753 high-risk counties in deep South, Appalachia, Mississippi Delta, and rural West regions of the United States. The top three most important predictors were premature death rate, childhood poverty rate, and percentage of rural population, which collectively contributed 70% of feature importance in the Random Forest analysis. This study aligns with larger epidemiological and social science literature and can inform intervention efforts in these areas.

In terms of Healthy and Safe Environments (mobile clinics, SNAP expansion, housing vouchers), Reducing Risky Behaviors (CDC DPP deployment, workforce health programs, health literacy outreach), and Clinical and Social Services (Medicaid expansion, FQHC funding, community health workers), the suggested intervention is in line with the CDC HI-5 three bucket framework. Every suggestion is

explicitly linked to a predictor found by the models, guaranteeing that intervention is supported by data rather than conjecture.

The data pipeline described in this report provides a template to make this one-time analysis sustainable public health surveillance tool. In refreshing the county risk scores each April based on latest data from the CDC and CHR, the framework allows for proactive resource allocation instead of reactive allocation, which is the promise of the HI-5 model.

The limitations of the analysis include an ecological fallacy of analysis, where results apply to the population level instead of the individual level, cross-sectional nature of the data, and lack of ACS data features that may capture additional dimensions of food insecurity and tenure types not captured by CHR data. Future work includes addition of longitudinal SDOH data, the effect of intersectionality on compounded risks such as poverty, rurality, and low educational level, and evaluation of outcomes in the 753 high-risk counties identified in analysis.

In the end, this framework shows how publicly accessible federal data, machine learning, and the CDC HI-5 evidence base can be combined to give county health officials and legislators a screening tool that is reproducible, annually updated, and equitably targeted, one that can allocate diabetes prevention resources to communities that most need them.

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